

# **SAGI Summer School 2025: Introduction to Fourier Analysis**

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# RC Circuit

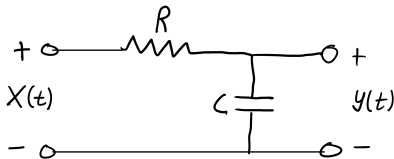
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- ① We will start our journey in Fourier analysis with some motivation from circuit theory. We will predict the time domain output of a circuit given an arbitrary input.
- ② The circuit we will consider is the simple RC circuit shown below, with a voltage signal  $x(t)$  applied to the input, and a voltage signal  $y(t)$  on the output:



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- ❶ Applying Kirchoff's voltage law, with  $I(t)$  being the current through the circuit:

$$x(t) = I(t)R + y(t) \quad (1)$$

- ❷ If  $Q(t)$  is the charge on the capacitor plate, we also have:

$$y(t) = \frac{Q(t)}{C} \quad (2)$$

$$I(t) = \frac{dQ(t)}{dt} \quad (3)$$

- ❸ Substituting, we get a differential equation describing the circuit behavior:

$$x(t) = RC \frac{dy(t)}{dt} + y(t) \quad (4)$$

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- ➊ Now suppose that the input is sinusoidal  $x(t) = e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$ .
- ➋ Let's test as a possible output solution a scaled version of the input:  
 $y(t) = Ae^{i\omega t}$
- ➌ Substituting these values into the differential equation:

$$e^{i\omega t} = RC \frac{d}{dt} (Ae^{i\omega t}) + Ae^{i\omega t} \quad (5)$$

$$= i\omega RCAe^{i\omega t} + Ae^{i\omega t} \quad (6)$$

$$= (1 + i\omega RC)Ae^{i\omega t} \quad (7)$$

- ➍ We see that:

$$y(t) = \left( \frac{1}{1 + i\omega RC} \right) x(t) = Ax(t) \quad (8)$$

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- ① Indeed, we see that the output is just a scaled version of the input!
- ② A fancy way to say this is that a complex exponential (sinusoid) is an **eigenfunction** of the differential operator describing the RC circuit.
- ③ Because the scaling of the input to the output  $A$  is frequency dependent, let's relabel:

$$A \rightarrow H(\omega) = \frac{1}{1 + i\omega RC} \quad (9)$$

so that:

$$y(t) = H(\omega)x(t) \quad (10)$$

- ④ Now, let's suppose that we take the input to be  $x(t) = e^{i\omega_1 t} + e^{i\omega_2 t}$ .  
Motivated by the previous result, let's guess that:

$$y(t) = H(\omega_1)e^{i\omega_1 t} + H(\omega_2)e^{i\omega_2 t} \quad (11)$$

$$= \left( \frac{1}{1 + i\omega_1 RC} \right) e^{i\omega_1 t} + \left( \frac{1}{1 + i\omega_2 RC} \right) e^{i\omega_2 t} \quad (12)$$

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- 1 Let's check if this  $y(t)$  is indeed a solution to eqn. (4).
- 2 Go back to the differential equation, substitute the input and the trial solution:

$$\begin{aligned} e^{i\omega_1 t} + e^{i\omega_2 t} &= \frac{i\omega_1 RC}{1 + i\omega_1 RC} e^{i\omega_1 t} + \frac{i\omega_2 RC}{1 + i\omega_2 RC} e^{i\omega_2 t} \\ &\quad + \frac{1}{1 + i\omega_1 RC} e^{i\omega_1 t} + \frac{1}{1 + i\omega_2 RC} e^{i\omega_2 t} \\ &= \left( \frac{1 + i\omega_1 RC}{1 + i\omega_1 RC} \right) e^{i\omega_1 t} + \left( \frac{1 + i\omega_2 RC}{1 + i\omega_2 RC} \right) e^{i\omega_2 t} \\ &= e^{i\omega_1 t} + e^{i\omega_2 t} \end{aligned}$$

- 3 Indeed, this  $y(t)$  is a solution. Emboldened by this result, let's generalize to an arbitrary summation of complex exponentials.

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## Statement

If the input signal to the RC circuit  $x(t)$  is an arbitrary summation of complex exponentials:

$$x(t) = \sum_{n=0}^{\infty} a_n e^{i\omega_n t} \quad (13)$$

Then the output signal  $y(t)$  of the circuit is:

$$y(t) = \sum_{n=0}^{\infty} H(\omega_n) a_n e^{i\omega_n t} \quad (14)$$

where:

$$H(\omega_n) = \frac{1}{1 + i\omega_n t} \quad (15)$$

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- ① This statement is actually quite general and applies to nearly every circuit that we will work with.
- ② It is really a statement that follows from linearity and time-invariance of the differential equation governing the circuit.
- ③ If we know the output of the circuit corresponding to a sinusoidal input with frequency  $\omega$ , then we know the output corresponding to some arbitrary summation of sinusoids of various frequencies.
- ④ This is because complex exponentials (sinusoids) are **eigenfunctions** of **linear-time-invariant** operators.



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- ① In the real physical world, we take the real part of complex exponentials for our input, and we use the **magnitude** and **phase** of  $H(\omega)$  to determine our output.
- ② That is,  $x(t) = e^{i\omega t} \rightarrow \mathcal{R}\{e^{i\omega t}\} = \cos(\omega t)$  and:

$$y(t) = H(\omega)x(t) \rightarrow |H(\omega)| \cos(\omega t + \angle H(\omega)) \quad (16)$$

where:

$$|H(\omega)| = \sqrt{\mathcal{R}\{H(\omega)\}^2 + \mathcal{I}\{H(\omega)\}^2} \quad (17)$$

$$\angle H(\omega) = \tan^{-1} \left( \frac{\mathcal{I}\{H(\omega)\}}{\mathcal{R}\{H(\omega)\}} \right) \quad (18)$$

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- ❶ For the RC circuit, we have:

$$H(\omega) = \frac{1}{1 + i\omega RC} = \frac{1 - i\omega RC}{1 + (\omega RC)^2} \quad (19)$$

- ❷ So we see that the real and imaginary components are:

$$\mathcal{R}\{H(\omega)\} = \frac{1}{1 + (\omega RC)^2} \quad (20)$$

$$\mathcal{I}\{H(\omega)\} = -\frac{\omega RC}{1 + (\omega RC)^2} \quad (21)$$

- ❸ And:

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \quad \angle H(\omega) = \tan^{-1}(-\omega RC) \quad (22)$$

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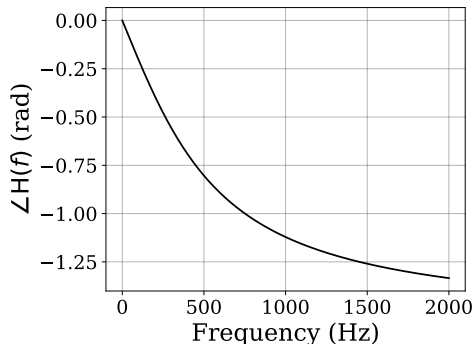
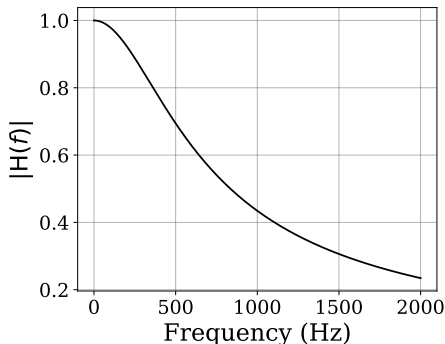
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- ❶ Let's make this a bit more concrete with a real circuit. let's take  $R = 1 \text{ k}\Omega$  and  $C = 330 \text{ nF}$ .
- ❷ Recalling that we have  $\omega = 2\pi f$ , we can plot the **magnitude** and **phase** response of the circuit:



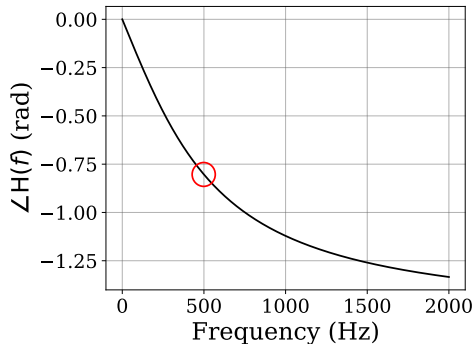
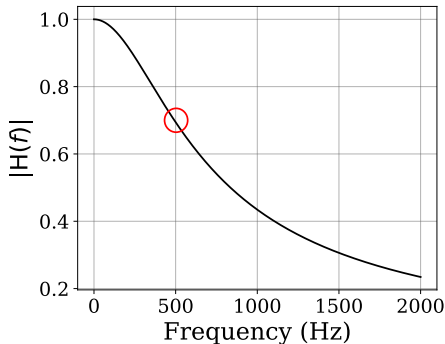
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- ① Taking  $f = 500$  Hz, we can predict the output by finding the correct points on the magnitude and phase plots:



- ② We predict roughly:

$$y(t) = 0.7 \cdot \cos(2\pi ft - 0.78) \quad (23)$$

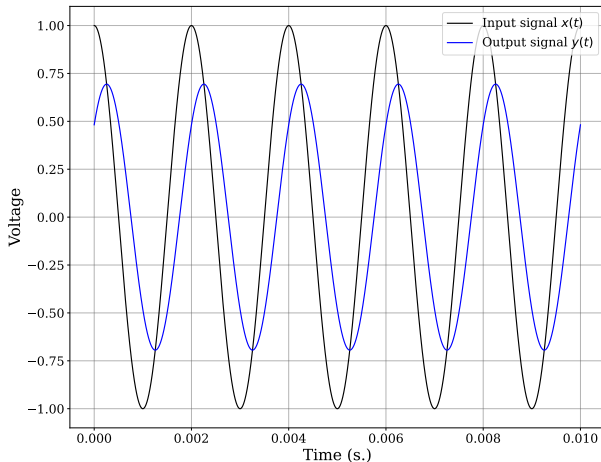
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① Plotting the input and predicted output together, we get:



② Let's compare this to a real circuit!

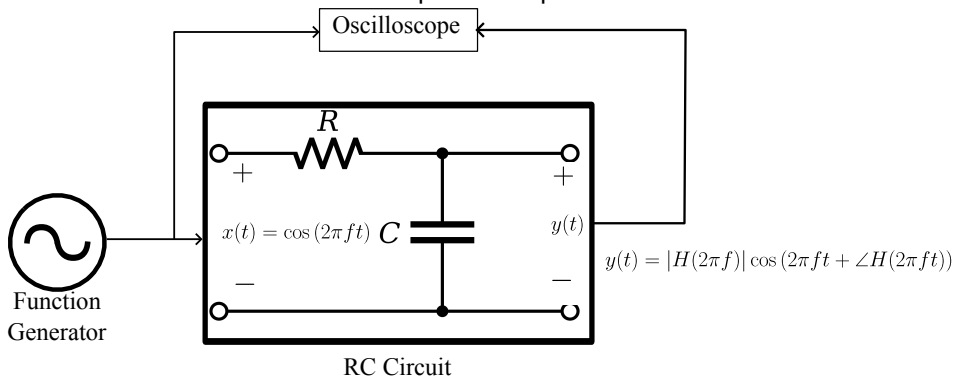
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- 1 To test our prediction, we need to build a real RC circuit. We also need a **function generator** and an **oscilloscope**.
- 2 The block diagram for the measurement is shown below. We use a function generator to create the sinusoidal input to the circuit, then we measure both the input and output of the circuit with an oscilloscope. We will compare the data measured with the oscilloscope to our prediction in a real circuit.



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- 1 A photo from the actual circuit demonstrated during the lesson is shown below:

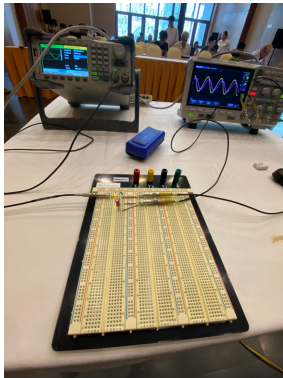


Figure: RC circuit live demo— In the top left is the function generator. Top right is the oscilloscope. In the center is the RC circuit constructed on a breadboard.

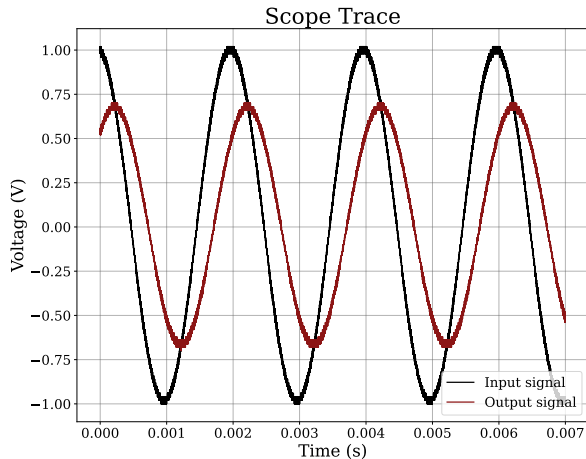
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- 1 Using the function generator to generate the input  $x(t) = \cos(2\pi ft)$ , with  $f = 500$  Hz, gives the following results:





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- 1 This looks very close to our prediction. Let's compare the output  $y(t)$ , the red data in the previous figure, directly with our predicted output signal. Plotting the data along with the predicted output we see:

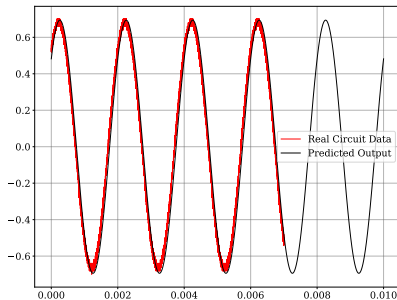


Figure: Predicted output plotted in black along with the real measured output in red. The y-axis is measured in volts, while the x-axis is in seconds.

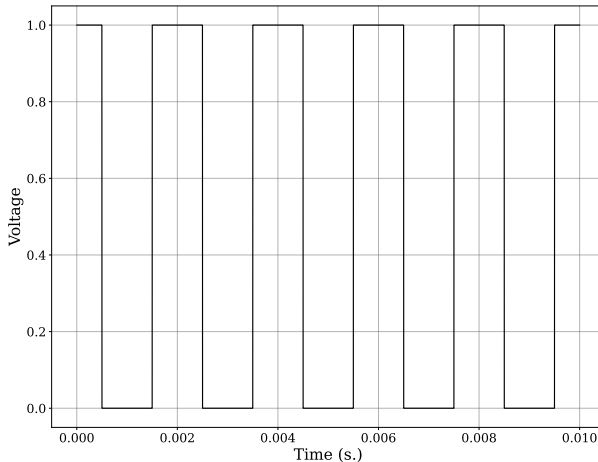
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- ① Now, let's try a slightly more complicated function to really test the bold statement of eqn. (14) above.
- ② Consider the square wave signal plotted below:



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- ① We will soon see that this signal can be represented as the following summation of sinusoids:

$$s(t) = \frac{1}{2} + 2 \sum_{n=0}^{\infty} \left[ \frac{2 \sin(n\pi/2)}{n\pi} \right] \cos(2\pi fnt) \quad (24)$$

- ② Using equation (14) and (16) above, we can predict the output:

$$y(t) = \frac{|H(0)|}{2} \cos(\angle H(0)) \quad (25)$$

$$+ 2 \sum_{n=0}^{\infty} H(2\pi fnt) \left[ \frac{2 \sin(n\pi/2)}{n\pi} \right] \cos(2\pi fnt + \angle H(2\pi fnt)) \quad (26)$$

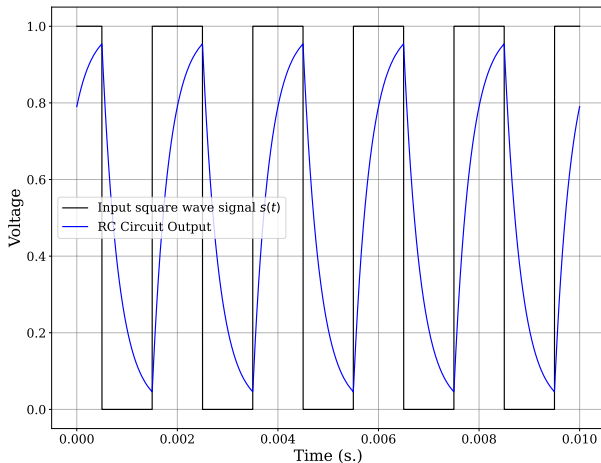
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- ① Using eqn. (25) and some python code, we predict the output of the circuit to be:



- ② Of course, we need to check this with an actual circuit!

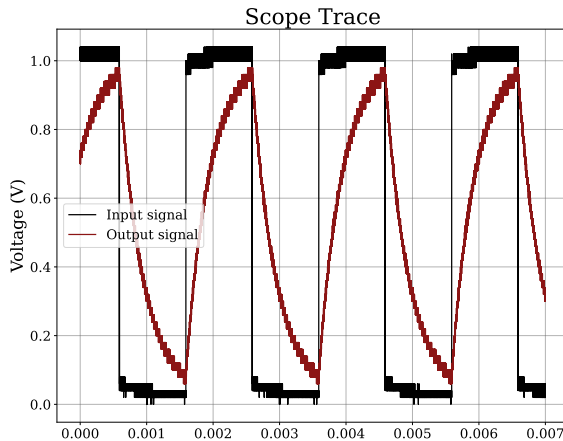
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- ① Using the same setup as for the sine-wave, we can generate an  $f = 500$  Hz square wave using the function generator. This time, the measured data taken by the oscilloscope reads:



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① Again plotting our predicted output directly against the measurement, we see:

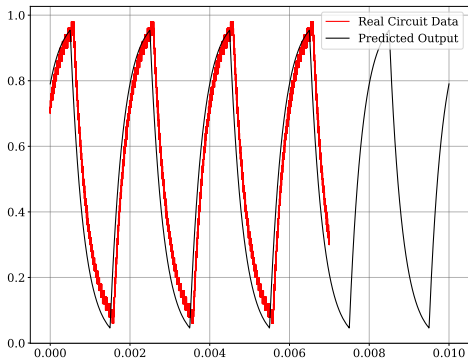


Figure: Here, the y-axis is in volts while the x-axis is seconds.

# RC Circuit: Summary

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- ➊ In this lecture we analyzed an RC low-pass filter circuit. We saw that the output of the circuit to a sinusoidal input is just a scaled and phase shifted version of the input.
- ➋ We extended this result to say that if the input to the circuit is any summation of sinusoids, the output will be scaled and phase shifted versions of these same sinusoids.
- ➌ We checked both of these results by predicting the output of the filter to a single sinusoid and to a square wave and we verified that our predictions match the behavior of the real circuit.
- ➍ Fourier analysis in circuit theory consists of treating all input signals as summations of sinusoids and designing circuits and systems to manipulate these sinusoids in a desired manner.